

NEXUS INSTALLATION

launch instance :

```
>> nexus_server
```

```
>> t2.medium
```

nexus port number : 8081

once check <free -h>

=====NEXUS INSTALLATION ON REDHAT=====

Step 1: Switch to Root User

```
>> sudo su -          #switch to root user
```

Step 2: Verify Java Installation

```
>> javac -version     #check whether java is installed or not
```

Step 3: Install Java

```
>> sudo rpm --import https://yum.corretto.aws/corretto.key
sudo curl -L -o /etc/yum.repos.d/corretto.repo https://yum.corretto.aws/corretto.repo
sudo yum install java-1.8.0-amazon-corretto-devel --nogpgcheck
```

Verify Java Version

```
>> java -version
```

Step 4: Install Required Packages

```
>> yum install tar wget tree -y
```

Step 5: Create Nexus User

```
>>sudo useradd nexus
```

Step 6: Move to /opt Directory

```
>> cd /opt
```

Step 7: Download Nexus

```
>> sudo wget https://download.sonatype.com/nexus/3/nexus-3.77.2-02-unix.tar.gz
```

Step 8: Extract Nexus Package

```
>> sudo tar -xvzf nexus-3.77.2-02-unix.tar.gz
```

Step 9: Rename Nexus Folder

```
>>sudo mv nexus-3.77.2-02 nexus
```

Step 10: Remove Tar File

```
>>rm -rf nexus-3.77.2-02-unix.tar.gz
```

Step 11: Give Sudo Permission to Nexus User

```
>>visudo
nexus ALL=(ALL) NOPASSWD: ALL
```

Step 12: Change Ownership and Permissions

```
>>sudo chown -R nexus:nexus /opt/nexus
sudo chown -R nexus:nexus /opt/sonatype-work
chmod -R 775 /opt/nexus
chmod -R 775 /opt/sonatype-work
```

Step 13: Configure Nexus User

```
>>sudo vi /opt/nexus/bin/nexus.rc
#run_as_user="nexus" remove comment
```

Step 14: Create Nexus Service File

```
>> sudo vi /etc/systemd/system/nexus.service
```

```
'''
```

```
[Unit]
```

```
Description=Nexus Repository Manager
```

```
After=network.target
```

```
[Service]
```

```
Type=forking
```

```
LimitNOFILE=65536
```

```
User=nexus
```

```
Group=nexus
```

```
ExecStart=/opt/nexus/bin/nexus start
```

```
ExecStop=/opt/nexus/bin/nexus stop
```

```
Restart=on-abort
```

```
[Install]
```

```
WantedBy=multi-user.target
```

Step 15: Switch to Nexus User

```
>> sudo su - nexus
```

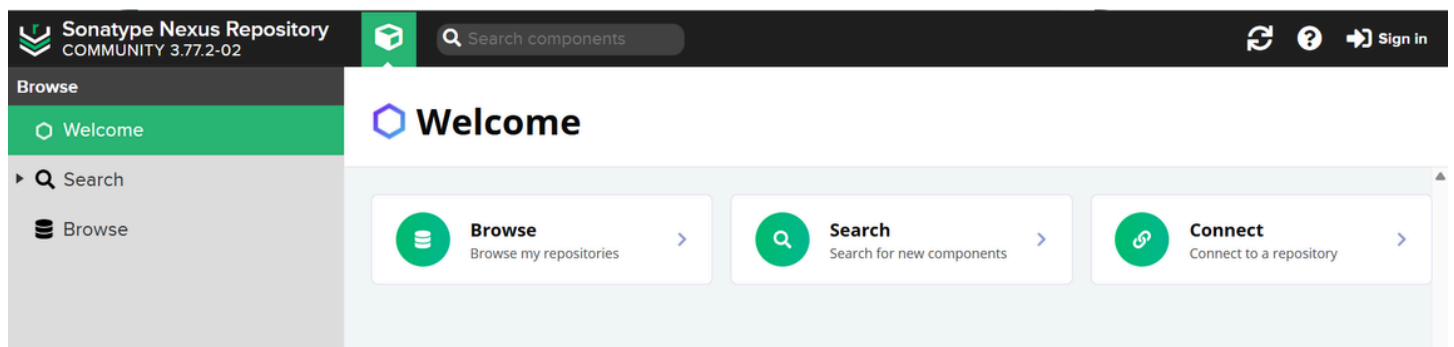
Step 16: Start Nexus Service

```
>> sudo systemctl start nexus
```

Step 17: Check Nexus Status

```
>>sudo systemctl status nexus
```

AFTER COMPLETE SETUP CHECK IP_ADDRESS:8081



CLICK ON SIGN IN AND USER NAME: admin AND PASSWORD WE CAN SET AGAIN LOGIN

Sign In

Your **admin** user password is located in
/opt/sonatype-work/nexus3/admin.password on the
server.

admin|

Password

Sign in

Cancel

TERMINAL : cat /opt/sonatype-work/nexus3/admin.password

LIKE THIS WE GET PASSWORD :

copy :

<< cbab532d-6b44-4ced-ba00-158ae742dba6 >>

PLEASE ENTER PASSWORD AND SAVE :

Please choose a password for the admin user

2 of 6

New password:

Confirm password:

! Passwords do not match

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ANONYMOUS ACCESS :

MOSTLY WE GIVE DISABLE ONLY

Configure Anonymous Access

5 of 6

Enable anonymous access means that by default, users can search, browse and download components from repositories without credentials. Please **consider the security implications for your organization**.

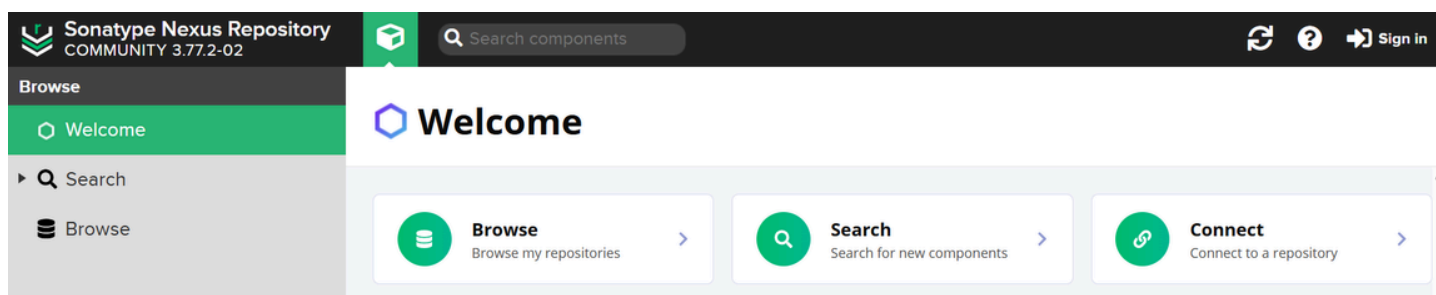
Disable anonymous access should be chosen with care, as it **will require credentials for all** users and/or build tools.

[More information](#)

- ☐ Enable anonymous access
- ☐ Disable anonymous access

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DISABLE :

Configure Anonymous Access

5 of 6

Enable anonymous access means that by default, users can search, browse and download components from repositories without credentials. Please **consider the security implications for your organization**.

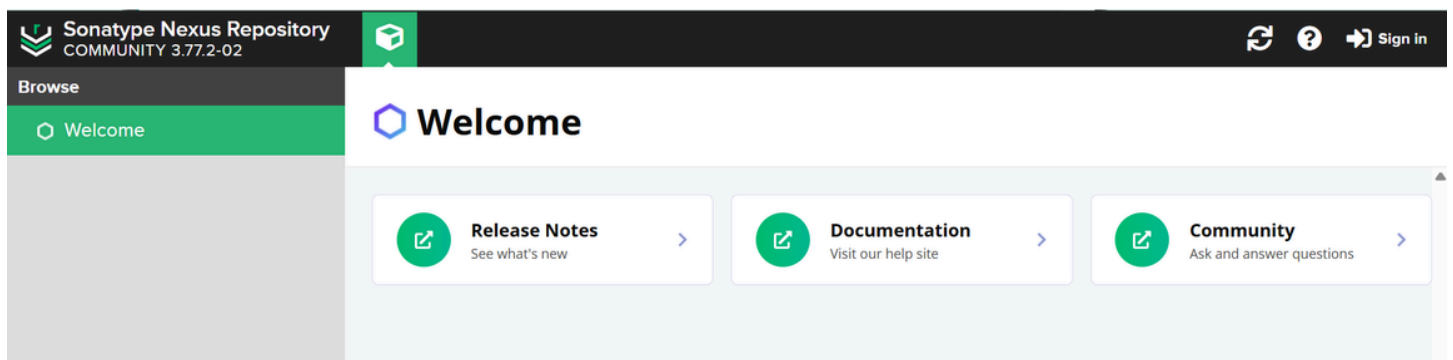
Disable anonymous access should be chosen with care, as it **will require credentials for all** users and/or build tools.

More information

- ☐ Enable anonymous access
- ☐ Disable anonymous access

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👉 “In Maven projects, Nexus repository URLs are configured in `pom.xml`, while authentication credentials like username and password are configured securely in `settings.xml`.”

using **Apache Maven** and Sonatype Nexus Repository:

- Nexus URL → usually configured in `pom.xml`
- Username/password → usually configured in `settings.xml`

Example `pom.xml` :

```
<project>

  <groupId>com.company</groupId>

  <artifactId>myapp</artifactId>

  <version>1.0</version>

  <distributionManagement>

    <repository>

      <id>nexus-releases</id>

      <url>

        http://54.158.95.243:8081/repository/maven-releases/

      </url>

    </repository>

    <snapshotRepository>

      <id>nexus-snapshots</id>

      <url>

        http://54.158.95.243:8081/repository/maven-snapshots/

      </url>

    </snapshotRepository>

  </distributionManagement>

</project>
```

=====

EXAMPLE settings.xml

cd ~/.m2/settings.xml

```
<settings>

  <servers>

    <server>

      <id>nexus-releases</id>
```

```
<username>admin</username>

<password>admin123</password>

</server>

<server>

  <id>nexus-snapshots</id>

  <username>admin</username>

  <password>admin123</password>

</server>

</servers>

</settings>
```

=====

HOW WE CAN CHANGE PORT AND CONTEXT PATH OF NEXUS

```
[nexus@ip-172-31-1-59 /]$ cd /opt/nexus/etc/
```

```
[nexus@ip-172-31-1-59 etc]$ ls
```

```
fabric jetty karaf logback nexus-default.properties ssl
```

```
vi nexus-default.properties
```

```
## DO NOT EDIT - CUSTOMIZATIONS BELONG IN $data-dir/etc/nexus.properties
```

```
##
```

```
# Jetty section
```

```
application-port=9091
```

```
application-host=0.0.0.0
```

```
nexus-args=${jetty.etc}/jetty.xml,${jetty.etc}/jetty-http.xml,${jetty.etc}/jetty-requestlog.xml
```

```
nexus-context-path=/bala
```

```
# Nexus section
```

```
nexus-edition=nexus-pro-edition
```

nexus-features=\

nexus-pro-feature

systemctl restart nexus

CHECK : <http://server-ip:9091/bala>

=====

HOW WE CAN CREATE REPOSITORY :

snapshot repository

>> go to repository and click create repo

>> select maven2<hosted>

The screenshot displays the Sonatype Nexus Repository Administration web interface. The top navigation bar includes the Sonatype logo, version 'COMMUNITY 3.77.2-02', a search bar, and user information 'admin' with a 'Sign out' link. The left sidebar shows the 'Administration' menu with 'Repositories' highlighted. The main content area is titled 'Create Repository: maven2 (hosted)' and contains the following form fields:

- Name:** A unique identifier for this repository. Value: project-snapshot
- Online:** ☒ If checked, the repository accepts incoming requests
- Maven 2**
 - Version policy:** What type of artifacts does this repository store? Value: Snapshot
 - Layout policy:** Validate that all paths are maven artifact or metadata paths. Value: Strict
 - Content Disposition:** Add Content-Disposition header as 'Attachment' to disable some content from being inline in a browser. Value: Inline
- Storage**
 - Blob store:** Blob store used to store repository contents. Value: default

release repository :

>> go to repository and click create repo

>> select maven2<hosted>

Sonatype Nexus Repository
COMMUNITY 3.77.2-02

Administration

- Repository
 - Repositories**
 - Blob Stores
 - Data Store
 - Proprietary Repositories
 - Content Selectors
 - Cleanup Policies
 - Routing Rules
- Security
 - Privileges
 - Roles
 - Users
 - Anonymous Access

Repositories / **Select Recipe** / **Create Repository: maven2 (hosted)**

Name: A unique identifier for this repository

Online: ☒ If checked, the repository accepts incoming requests

Maven 2

Version policy:
 What type of artifacts does this repository store?

Layout policy:
 Validate that all paths are maven artifact or metadata paths

Content Disposition:
 Add Content-Disposition header as 'Attachment' to disable some content from being inline in a browser.

Storage

Blob store:
 Blob store used to store repository contents

COPY URL :

release url :

<http://65.2.150.21:8081/repository/project-release/>

snapshot url :

<http://65.2.150.21:8081/repository/project-snapshot/>

we should update in pom.xml

Example pom.xml :

```
<project>
```

```
<groupId>com.company</groupId>
```

```
<artifactId>myapp</artifactId>
```

```
<version>1.0</version>
```

```
<distributionManagement>
```

```
<repository>
```

```
<id>nexus-releases</id>
```

```
<url>
```

```
http://65.2.150.21:8081/repository/project-release/
```

```
</url>
```

```
</repository>

<snapshotRepository>

  <id>nexus-snapshots</id>

  <url> http://65.2.150.21:8081/repository/project-snapshot/ </url>

</snapshotRepository>

</distributionManagement>

</project>
```

```
=====
RUN GOAL : upload artifact to nexus
mvn clean deploy
=====
```

DIFFERENCE BETWEEN SNAPSHOT AND RELEASE :

snapshot :

```
>>snapshot are used for Development/testing env
>> snapshot will allowing redeploy
<project>
```

```
<modelVersion>4.0.0</modelVersion>
```

```
<groupId>com.company</groupId>
```

```
<artifactId>myapp</artifactId>
```

```
<version>1.0-SNAPSHOT</version>
```

```
</project>
```

release :

```
>>release are used for production env
>> release will not allowing redeploy
```

```
<project>
```

```
<modelVersion>4.0.0</modelVersion>
```

```
<groupId>com.company</groupId>
```

<artifactId>myapp</artifactId>

<version>1.0</version>

</project>

=====

Role :

Create Role

Role Type

Type *

Nexus role ▼

Role Setup

Role ID *

devops-role ✓

Role Name *

DevOps Team Role ✓

Role Description

Role for DevOps engineers to upload and manage Maven artifacts ✓

Applied Privileges

[Modify Applied Privileges](#)

Applied Roles

[Modify Applied Roles](#)CancelSave

Read Access : nx-repository-view-maven2-*-read

Meaning:

👉 read all Maven repositories.

Upload Access : nx-repository-view-maven2-*-add

Meaning:

👉 upload artifacts.

Edit Access : nx-repository-view-maven2-*-edit

Meaning:

👉 modify repository contents.

Delete Privilege : nx-repository-view-maven2-*-delete

Meaning:

👉 delete artifacts from Maven repositories.

USERS :

 **Users**

/  **Create User**

ID:
This will be used as the username

First name:

Last name:

Email:
Used for notifications

Password:

Confirm password:

Status:

Roles:

Available



nx-admin

nx-anonymous

>

<

Granted

DevOps Team Role

Create local user

Cancel

LOGIN WITH NEW USER WE CAN SEE LIKE THIS :

CLEAN POLICIES :

Create Cleanup Policy

Define Cleanup Policy

Cleanup policies allow you to automatically remove unused components and free up storage space in your Nexus repository instance. You can schedule cleanup policies to run at specific intervals and define cleanup criteria for selecting components to delete.

Name *

Use a unique name for the cleanup policy.

snapshot-cleanup

Format *

The format that this cleanup policy can be applied to.

maven2

Description

Delete old snapshot artifacts after 30 days.

Release Type

Remove components that are of the following release type.

Releases & Pre-Releases(Snapshots)

Cleanup Criteria *

Remove all components that match all selected criteria. At least one criterion must be selected.

☒ Component Age *

Components published over "X" days ago (eg 1,000)

30

☒ Component Usage *

Components downloaded in "X" amount of days (eg 1,000)

100

☒ Asset Name Matcher *

Remove components that have at least one asset name matching the following regular expression pattern.

SNAPSHOT

Cancel

Save

 **Cleanup Policies**

Create Cleanup Policy

Manage component removal configuration

Filter

NAME	FORMAT	
snapshot-cleanup	maven2	Delete old snapshot artifacts after 30 days

What is a cleanup policy?

Cleanup policies can be used to remove content from your repositories. These policies will execute at the configured frequency. Once created, a cleanup policy must be assigned to a repository from [the repository configuration screen](#). For more information, check [the documentation](#).